

TARUN MURALI

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EDUCATION

University of Maryland

B.S. Computer Science, Minors in Cybersecurity (Honors) and Statistics.

Computer Science: Computer Vision, Operating Systems, Networks, Compilers, Penetration Testing, Parallel Computing

Math: Cryptography, Algorithms, Abstract Algebra, Linear Algebra, Statistics I/II, Calculus

College Park, MD

Aug. 2020 – May. 2024

EXPERIENCE

CoStar Group

Software Engineer

Jul. 2024 – May. 2025

Washington, D.C.

- Built a high-throughput secrets management service with automated failover, ensuring zero downtime for 10+ teams.
- Cut infra costs 5× with a **Python** proxy API that caches secrets and optimizes fetches by developer access patterns.
- Created an authentication gateway in **TypeScript** for secure, role-based access to **Azure** resources.
- Built a Terraform module deploying AzureAI infrastructure, supporting high-scale ML workflows on Homes.com.

CoStar Group (Apartments.com)

Software Engineering Intern

Jun. 2023 – Aug. 2023

Washington, D.C.

- Reduced query latency by 70% by building a **JSON-to-SQL** transpiler for large-scale property data.
- Integrated language tool into Node pipelines through **Rust** interpreter compiled to **WebAssembly (WASM)**.
- Automated Pod autoscaling for **AWS SQS** and **Kafka** triggers with **Terraform**: improved app response times by 3x.

Liberty Mutual Insurance

Software Engineering Intern

May. 2022 – August. 2022

Remote

- Built a **React.js** frontend for deployment image tracking, enabling ID powered searches.
- Created a Flask REST API for analyzing Splunk and Datadog logs to monitor pod crashes and uptime.
- Curtailed build pipeline traversals via **Redis** caching: Resulted in 55% speedup of search results.

NASA Harvest Lab @ UMD

Machine Learning Researcher

Jan 2022 – Aug. 2022

College Park, MD

- Prototyped crop forecasting model to isolate high-signal land regions. Fine-tuned **Extra Trees/Random Forest** models.
- Produced harvest quality control filters on global scale. Trained **Random Forest Regressor** on geospatial data.

PROJECTS & AWARDS

MessageMe | Java, Spring-Boot, Angular, Hibernate, Liquibase

May. 2025

- Developed a real-time messaging application using Spring Boot with WebSocket support
- Integrated PostgreSQL with JPA/Hibernate for efficient data persistence and Liquibase for schema migrations
- Implemented user authentication, message history retrieval, and chat room logic

GeekOS | C

Aug. 2023 – Dec. 2023

- Implemented core OS primitives like pipes, virtual memory, signals, and a distributed file system on a Linux kernel variant
- Focused on process and memory management, including paging-based isolation

Detoureee | Networks, Geo-Spatial Computing, Next.js, Cockroach DB

Apr. 2023

- Calculates quickest route around real-time map disruptions. **Routing Algorithm** optimizes shortest path/safety.
- Caches updated real-time routes. Users can input, update, delete disruptions by drawing polygons on map.

AutoViewAi | Next.js, Firebase, Vapi, React.js

May. 2025

- Platform to schedule and manage interviews with AI voice bots for automated candidate assessments.
- Delivers analysis of conversation dynamics with actionable speech insights, aid for visual and auditory impairments.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, C, Go, OCaml, Ruby, SQL, Rust

Web Technologies and Frameworks: HTML, CSS, JavaScript, React, Angular, Flask, Spring-Boot, Next.js, Langchain

Databases: MySQL, PostgreSQL, MongoDB, Pinecone, Weaviate, ChromaDB

Cloud and DevOps: Kubernetes, AWS, Azure, AzureDevOps, Docker, Terraform, Git, Linux, YAML

Libraries: pandas, NumPy, Matplotlib, TensorFlow

Certifications: AWS Certified Solutions Architect – Associate (2025)